Narendra Kumar Rayavarapu

DATA ENGINEER / ANALYST

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SUMMARY

Results-driven Data Engineer/Analyst with expertise in cloud-native data solutions across AWS, Azure, and Snowflake. Proficient in designing and optimizing scalable ETL/ELT pipelines using Databricks, PySpark, and Informatica PowerCenter. Skilled in big data processing, distributed computing, and machine learning to derive actionable insights. Experienced in Kafka-based streaming pipelines, CI/CD deployment for data workflows, and real-time analytics solutions. Certified in AWS, Microsoft Azure, and Tableau. Passionate about optimizing data pipelines, automating workflows, and delivering high-performance data architectures to support decision-making.

EDUCATION

University of South Dakota, South Dakota - USA Aug 2023 – May 2025
Master of Science: Computer and Information Science, GPA: 3.89/4.0.
Vignan'S Lara Institute of Technology and Science, Guntur - India July 2018 – May 2022
Bachelor of Technology in Computer Science, CGPA: 8.4/10.0

TECHNICAL SKILLS

Programming: Python, SQL, Java, C++, C, Scala, Shell Scripting
Big Data & ETL: PySpark, Snowflake, Apache Kafka, Hadoop, Informatica PowerCenter
Machine Learning & AI: Scikit-learn, TensorFlow, PyTorch, Predictive Modeling, NLP
Visualization: Tableau, Power BI, Amazon QuickSight
Cloud Platforms: AWS (S3, Glue, Lambda, EC2, Redshift, CloudWatch), Azure (Databricks, Synapse, Data Factory)
DevOps & Automation: Terraform, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD Pipelines
Databases: Oracle, PostgreSQL, MongoDB, NoSQL, MySQL

WORK EXPERIENCE

Pan Fed Credit union– Sr.Data Engineer May 2024 – April 2025

- Led the design and implementation of enterprise-scale ETL pipelines using AWS Glue and Snowflake to streamline the ingestion and transformation of high-volume financial datasets
- Developed event-driven data ingestion frameworks leveraging AWS Lambda and Step Functions, enabling responsive and near real-time data integration from core banking systems
- Constructed performance-optimized Snowflake schemas and refined SQL logic to reduce query latency and support dynamic, business-critical analytics workflows
- Created interactive Power BI dashboards powered by DAX, providing business stakeholders with real-time insights into customer trends, product usage and KPIs
- Established end-to-end monitoring and automated alerting for ETL pipelines using CloudWatch metrics, custom logs, and SNS notifications, ensuring system resilience and reliability
- Applied rigorous cloud security practices including role-based access, encryption standards, and audit logging to meet internal compliance and regulatory requirements.

Wartsila – Data Engineer Sep 2022 – July 2023

- Designed and implemented high-performance ETL workflows using Informatica PowerCenter, integrating data from SAP, Salesforce, and Oracle systems
- Automated job scheduling and monitoring with BMC Control-M, significantly reducing manual effort and ensuring SLA adherence
- Optimized database performance through partitioning, indexing, and materialized views, improving query efficiency by up to 50%
- Developed complex Oracle database objects and optimized ETL mappings for SCD Type 1 and Type 2 implementations
- Delivered executive dashboards and predictive analytics using Power BI and Tableau, supporting real-time and historical business insights

- Implemented real-time data streaming with Kafka and Snowflake to enhance data availability and reduce latency in critical pipelines
- Collaborated with cross-functional teams to manage end-to-end production data pipelines and align technical solutions with business needs.

Punjab National Bank – Informatica / ETL Developer Jan 2020 – Sep 2022

- Designed and deployed scalable ETL pipelines in Azure Data Factory to ingest and transform high-volume financial data from on-premises and cloud systems
- Developed PySpark-based data transformation jobs in Azure Databricks to process transaction logs, customer data, and credit risk indicators
- Orchestrated complex data workflows using Apache Airflow and ADF pipelines to ensure timely and reliable data delivery across departments
- Integrated third-party APIs and legacy banking systems to centralize operational data into Azure SQL Database and Azure Synapse Analytics
- Optimized storage and performance by implementing partitioning and dynamic data loading strategies in data lakes and Azure Synapse
- Built monitoring dashboards and implemented alerting for pipeline failures using Azure Monitor and Log Analytics
- Participated in Agile sprints, collaborating with data engineers, analysts, and risk management teams to meet business goals requirements

PROJECTS

Real -Time AI-Powered Analytics Platform on Google Cloud

- Developed a real-time data analytics solution using GCP services, streaming data from Pub/Sub into Big Query for instant insights and decision-making.
- Automated ETL workflows with Cloud Dataflow and Cloud Composer to process diverse datasets, including transactional and user behavior logs.
- Integrated AI models using Vertex AI for predictive analytics, such as churn and anomaly detection, with outputs visualized in Looker Studio dashboards.
- Ensured secure data handling with Cloud Storage, IAM policies, and end-to-end monitoring to support compliance and operational efficiency.

Customer Churn Prediction and Insights Pipeline on Azure Databricks

- Developed an end-to-end data pipeline on Azure Databricks using PySpark for cleaning, transforming, and analyzing customer behavior data from Azure Data Lake.
- Trained and deployed ML models using Spark MLlib to predict churn likelihood, integrating results with Power BI.
- Enabled real-time insights with Azure Synapse integration and implemented experiment tracking and model evaluation using MLflow within Databricks.

CERTIFICATIONS

- AWS certified Solution Architect - Associate (AWS).
- Google Data Analytics Professional Certificate (Google).
- Joy of computing using Python (NPTEL).
- Microsoft Azure Cloud (Simplilearn).
- Tableau Desktop Development (Simplilearn).
- Data Analytics using Python,SQL and Excel (Cisco).

RESEARCH

- Published research paper titled "A Comparative Study on Fake News Detection Using CNN-LSTM Models" in the Journals of Composition Theory. Conducted extensive research and data analysis to validate the findings using real-world datasets.